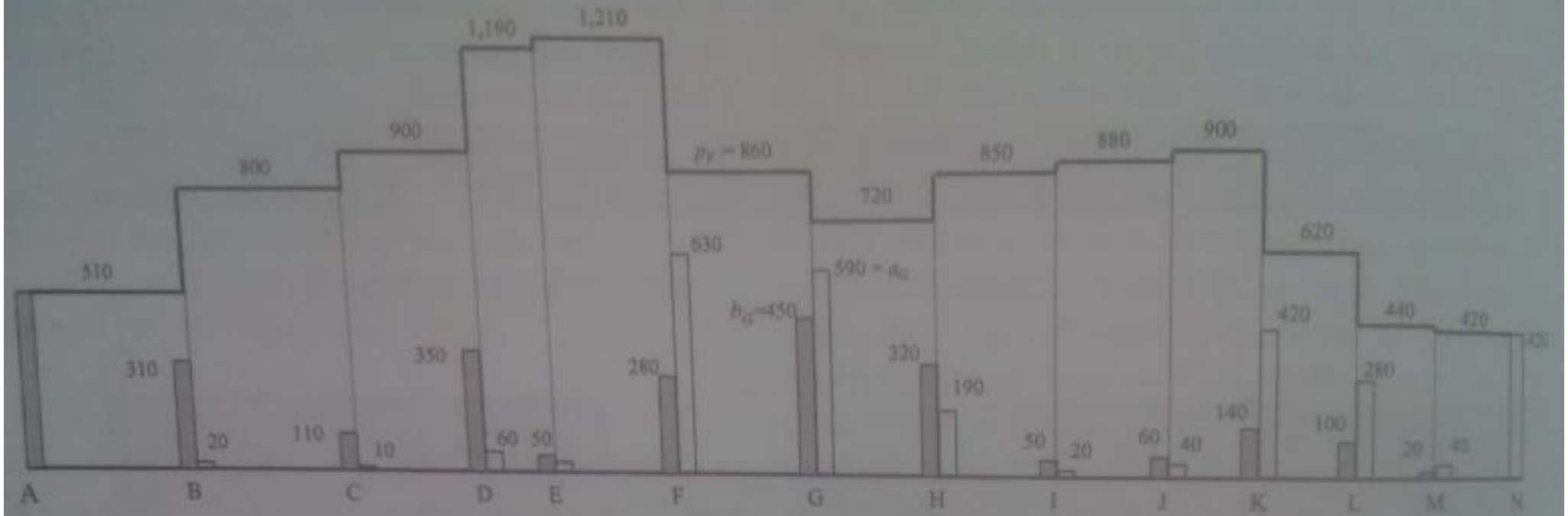
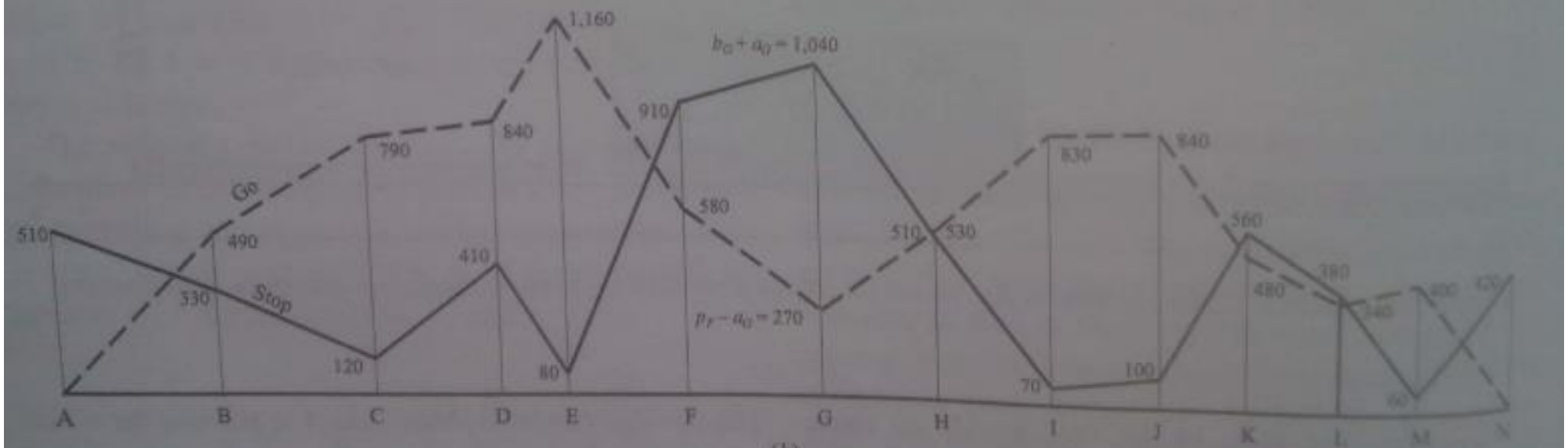
LECTURE 11

Application of Theoretical Analysis to Rapid Transit Station Planning

- Need for a station at any point along a rapid transit line depends on the ratio of two groups of people: desiring to board or alight- want to have station; while traveling further do not want to have any station
- Planning of station locations could start by tentatively designating a large number of potential locations
- At each station the sum of boarding and alighting passengers is plotted and a line is drawn connecting those values, called the ***Stop Line***
- Number of persons traveling on trains minus those who want to alight, represents those who do not want to be delayed by a station
- ***Go Line*** connects those values at stations for visual clarity
- Stop Value has a considerably greater relative weight than the Go Value at a station
- Only stations at which Stop is drastically lower than Go should be eliminated


1,190
1,210
900
800
510
$p_F = 860$
720
850
880
900
620
630
590 = a_G
b_G=450
420
440
420
420
350
280
320
190
310
280
140
100
110
60
40
50
20
20
60
50
20
10
A
B
C
D
E
F
G
H
I
J
K
L
M
N

(a)


1,160
$b_G + a_G = 1,040$
910
840
790
830
840
Go
580
510
490
510
530
560
410
380
530
Stop
480
$p_F - a_G = 270$
120
80
70
100
A
B
C
D
E
F
G
H
I
J
K
L
M
N

(b)